

Lab 10: Electrical Schematics

AE 8883 Spacecraft Engineering

Summary:

For this assignment, you will be creating a simple electrical schematic in Autodesk EAGLE. This will involve using existing default component libraries, as well as an imported component library we will provide. You will then add parts to the schematic, connect relevant pins with nets, and learn how to name nets for organization. Once the schematic is finished, you will learn how to have EAGLE perform an ERC for you, and then generate a BOM from your circuit.

Preparation: (EAGLE Installation)

1. Login with an Autodesk account with Educational Access, or create one by navigating to the following link and hitting **Get Started**:
<https://www.autodesk.com/education/edu-software/overview>
 - a. Note: If you aren't able to get Educational access for your account approved by Monday, don't worry, the features available in the free version of EAGLE will still allow you to do all of the lab
 - b. More information on obtaining Educational Access can be found here:
<https://knowledge.autodesk.com/customer-service/account-management/autodesk-for-education/students-and-educators/get-started-for-students-and-educators#confirm-eligibility>
2. Download the EAGLE installer for your preferred operating system here:
<https://www.autodesk.com/products/eagle/free-download>

Note for Windows 11 users!

If you are doing a clean install of EAGLE, you will need to make the following modification after installation to get it to open:

1. Go to your EAGLE installation folder
E.g. **C:\EAGLE9.6.2**
2. Rename **LIBEAY32.DLL** to **LIBEAY32.DLL.BAK** (make sure the file extension gets modified)

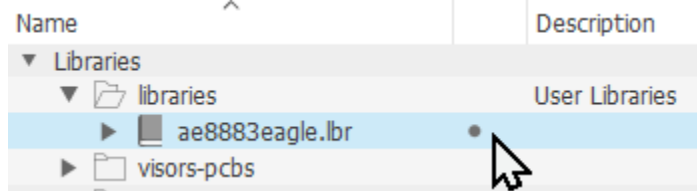
Application:

EAGLE Project Setup

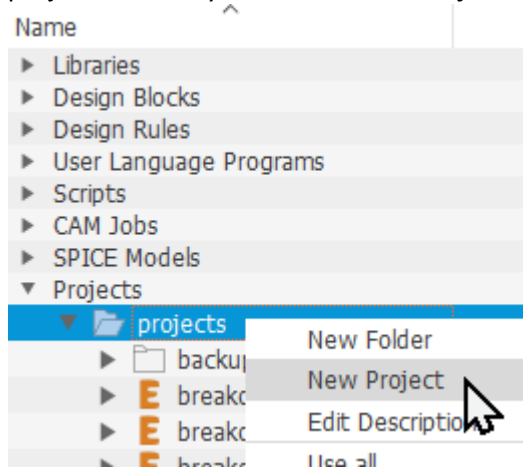
Upon starting EAGLE, you should have a Control Panel window open.

1. Import the provided **ae8883eagle.lbr** EAGLE component library that accompanies this lab. It can be downloaded from the Electrical Schematics Canvas Module
 - a. After downloading the component library, place it in a directory you're comfortable with, and **copy the filepath** to that directory.
 - b. In the **Control Panel** window, select **Options > Directories**.
 - c. In the **Libraries** text field, go to the end of the text currently inside it, add a semicolon as a separator, and then paste the filepath to the directory containing the downloaded component library. Do not delete the filepaths already in the text field.
 - d. Click OK to save your changes.

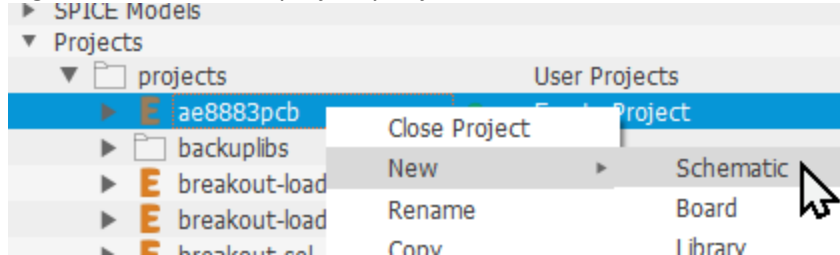
- In the **Control Panel**, expanding the **Libraries > libraries** nested drop-down should now display the ae8883eagle.lbr component in its list.



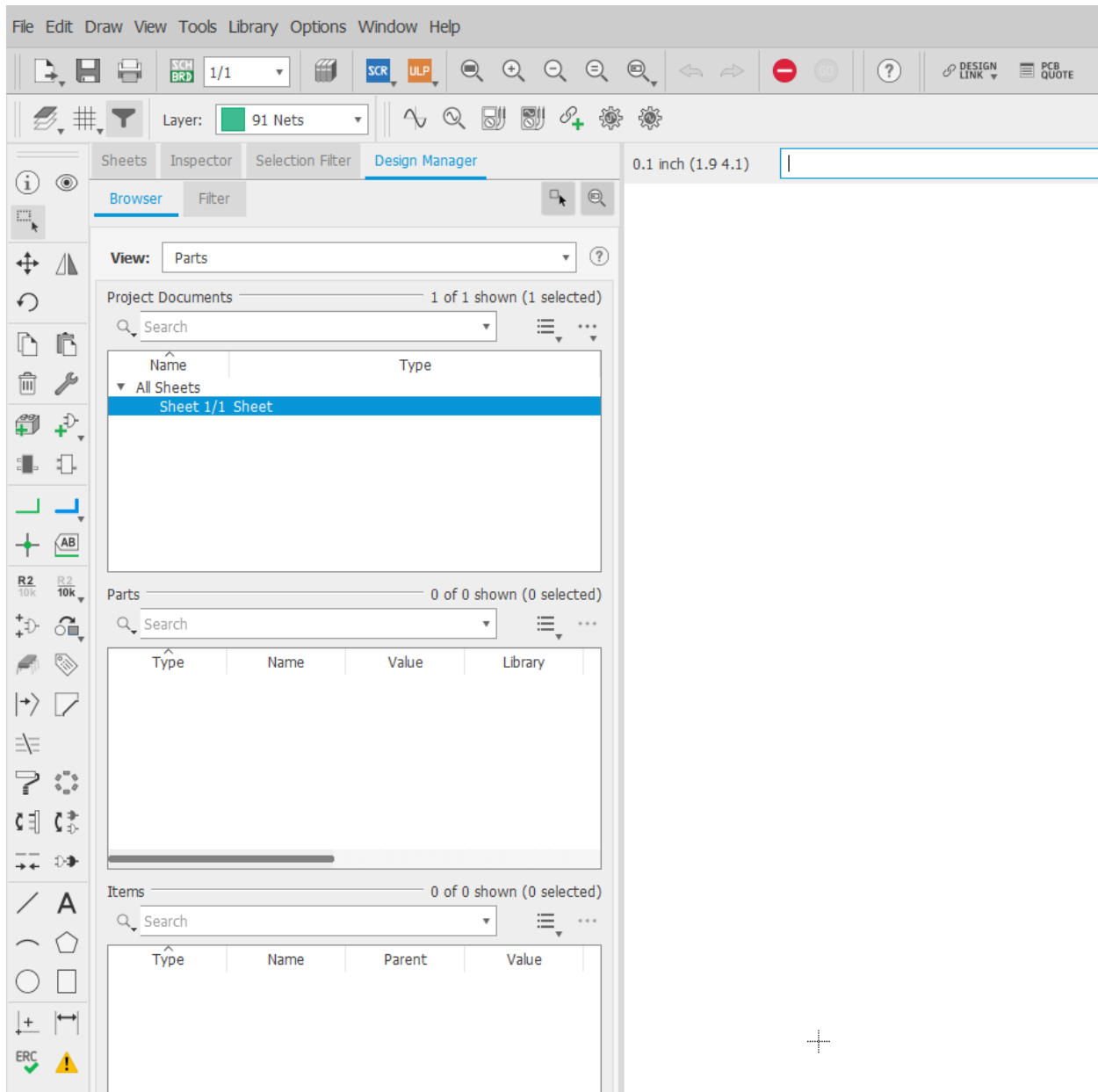
- Tell EAGLE that you want to **Use** this library for projects you're working on this session by clicking on the gray dot to turn it green. Alternatively, you can right-click the library and click **Use** to check it and achieve the same result. In EAGLE, all imported libraries must be included in projects in this manner for them to show up in the part selection menu.
- In the **Control Panel**, expand the **Projects > projects** nested drop-down, right click on the projects directory and select **New Project**. Title the project **ae8883pcb**, or whatever you'd like.



- Right-click on the new project you just created, and select **New > Schematic**



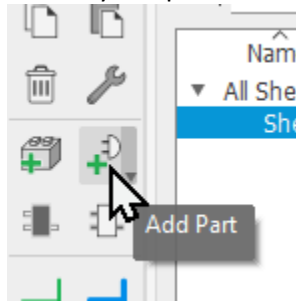
6. A new **Schematic** window should now have opened up. It should look similar to the following:



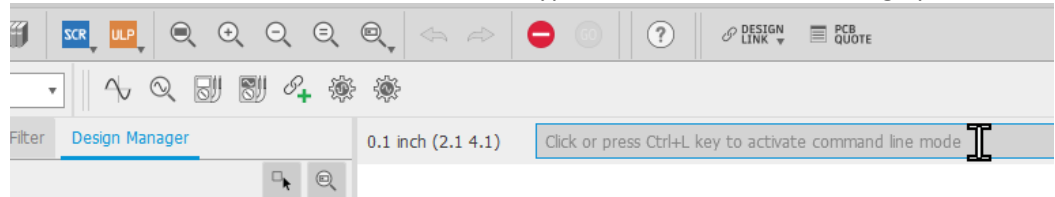
EAGLE Schematic Creation

1. First thing to do is save this schematic. Do so often as you work on it. You're free to give it the same name as the project, **ae8883pcb.sch**
2. We'll start by placing a frame down. This frame contains title block information that we'll want to fill out to document our project. To add this frame and other components to our schematic, open up the **Add Part** menu.

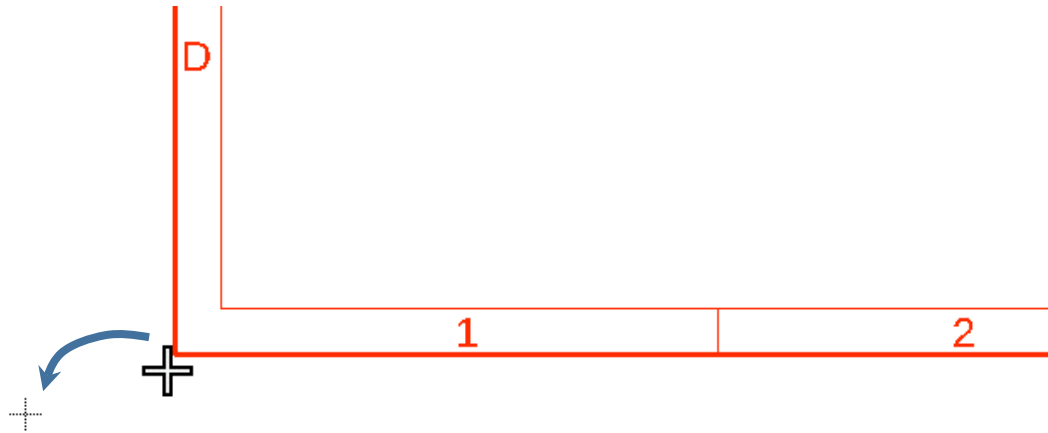
- One way to open this menu is to click on the Add Part icon in the sidebar.



- The other way is to use the EAGLE command line. You can either click it or press **Ctrl+L** to make it the active text field. Afterwards, type **add** and hit Enter to bring up the menu.



- Find the **ae8883eagle** library that we imported earlier in the list, expand it, and select the **A4L-LOC** frame. If you can't find it, you might have messed up the import process, try hitting F5 to refresh the Control Panel menu.
- Place the frame onto the schematic. Try to align the bottom-left of the frame with the gray crosshair denoting the "origin" of the schematic.



- After clicking once to place the frame down, press **ESC** to exit part placement for your selected component. For now, you can close the Add Part menu.
- Before we proceed, let's fill out the frame for documentation. Right-click an interactable part of the frame and select **Attribute**.
 - Interactable portions of a schematic symbol in EAGLE are marked with a tiny red crosshair.

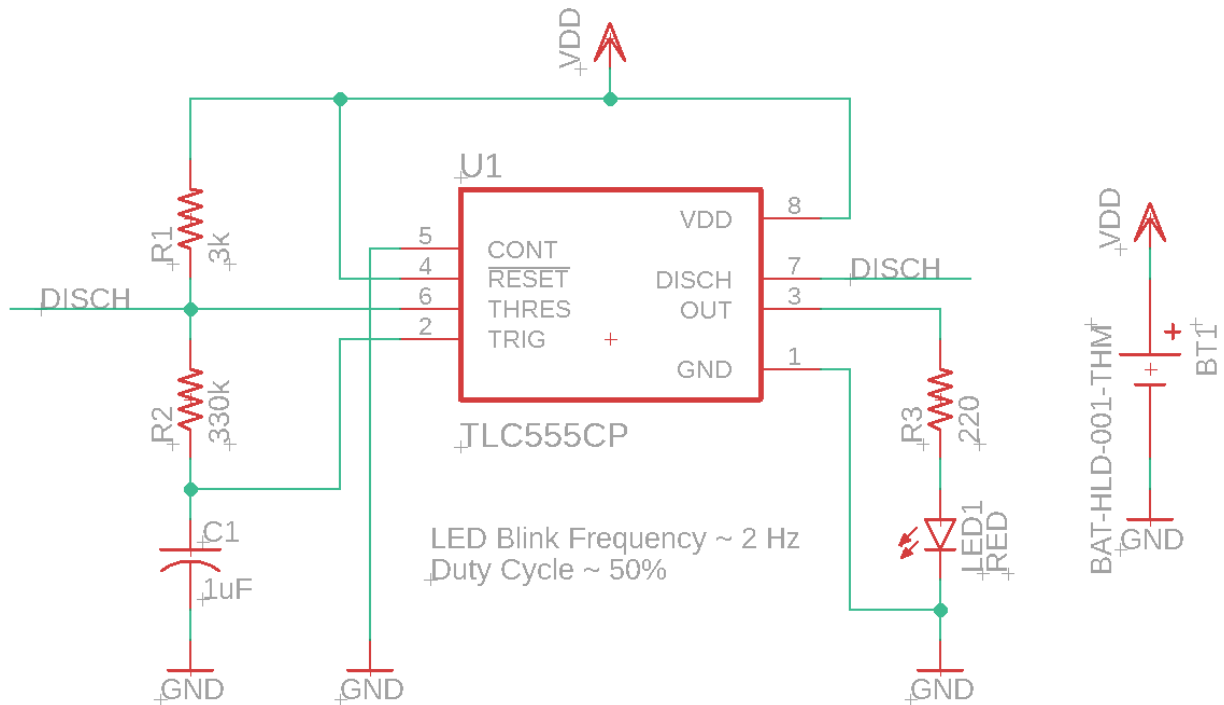
- On the frame we're using, this includes the bottom-left corner of the frame (you'll need to zoom in if you want to see it), or the bottom-left portions of any of the text fields on the frame that are already filled out, such as the Project Name.

Project: AE 8883 Spacecraft Eng I		D	
Georgia Tech Space Systems Design Laboratory		Designer:	
		Date:	
Sheet: 11		Rev: 10	
4	5	6	

- Fill out the DESIGNER_NAME (yourself), SCH_NAME (schematic name, E.g. **PCB Lab**), and DATE with the relevant information and click OK to save.

This specific frame is sized for an A4 sheet of paper. We'll want to keep any circuit symbols, nets, and notes within this frame.

This is the circuit you'll be replicating during this lab. It is an astable 555 timer circuit that blinks an LED.

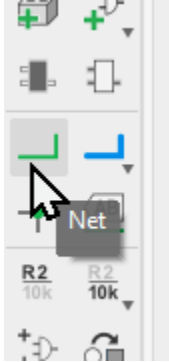


All parts can be found by searching for the following in the Add Parts menu:

- The **VDD** and **GND** supply symbols can be found in the default **supply1** library.
- The LED diode is the **LED5MM** diode that can be found in the default **led** library.
- The resistors are all through-hole R-US_0204/7 resistors in the default **rc1** library.
- The capacitor is a through-hole C-US050-024X044 capacitor in the default **rc1** library.
- BAT-HLD-001-THM (a button battery holder) and TLC555CP (a low power 555 timer) can be found in the **ae8883eagle.lbr** library we imported earlier.

Start off by first adding these components to the schematic. Once everything is on the schematic, we can start connecting them with nets.

The Net tool can be activated by clicking on the **Net** icon in the sidebar, or by typing **net** into the command line. Press ESC to cancel out of the tool and reenter Selection mode once you're finished.



If you feel that certain nets are annoying to route around components on the schematic, you may choose to name nets instead to reduce clutter. You can see I chose to do this with the DISCH signal. The VDD and GND supply symbols also serve this purpose; they aren't interpreted as individual devices by EAGLE, so we can use them to connect multiple items to both power and ground.

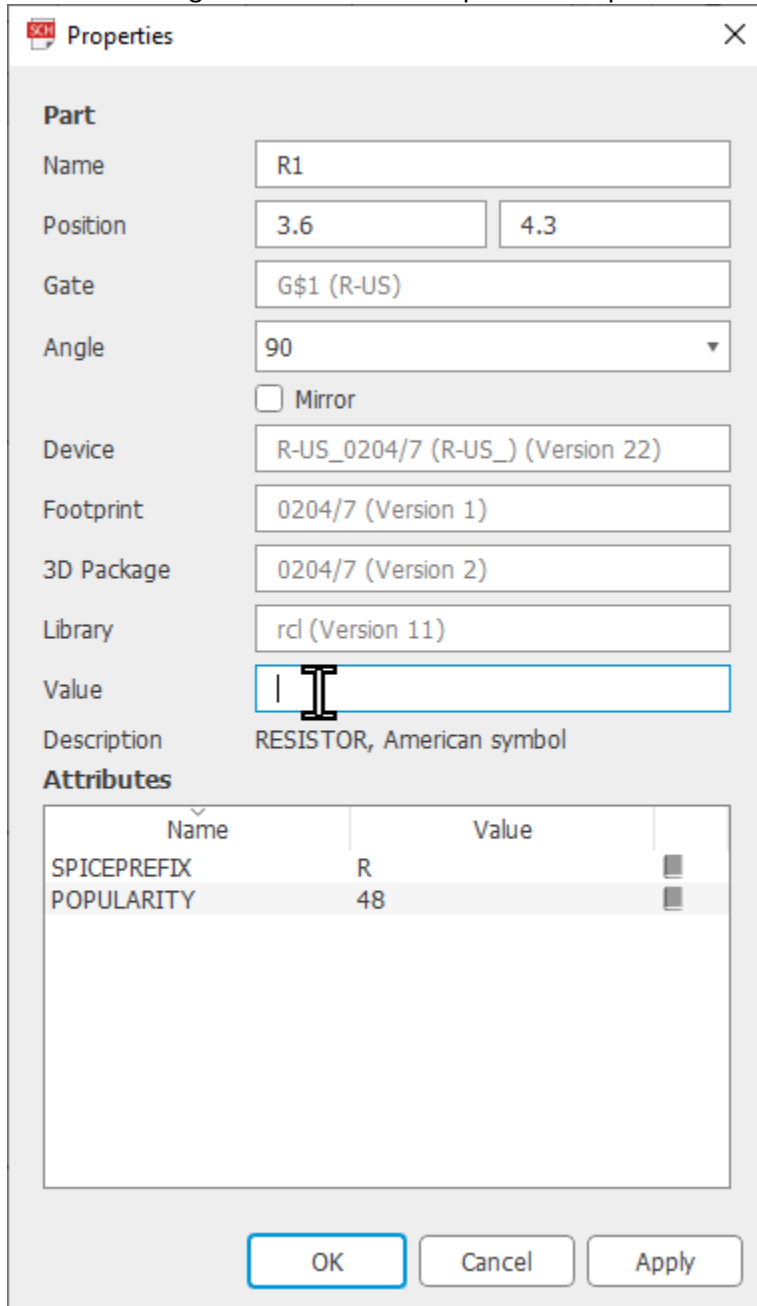
The Name tool can be activated by clicking on the **Name** icon in the sidebar, or by typing **name** into the command line. Press ESC to cancel out of the tool and reenter Selection mode once you're finished.



For the purposes of naming Nets, you'll want to use the Name tool instead of the Label tool. **The Label tool does not allow you to label separate nets with the same name to connect them, so don't use it.**



Certain components in our circuit (the resistors and capacitor) need values specified for them, so we'll do that now. Right-click each of those passive components and select **Properties**.



The screenshot shows the 'Properties' dialog box for a component. The 'Part' section contains the following fields:

- Name: R1
- Position: 3.6, 4.3
- Gate: G\$1 (R-US)
- Angle: 90
- ☐ Mirror
- Device: R-US_0204/7 (R-US_) (Version 22)
- Footprint: 0204/7 (Version 1)
- 3D Package: 0204/7 (Version 2)
- Library: rcl (Version 11)
- Value: | (with a cursor)
- Description: RESISTOR, American symbol

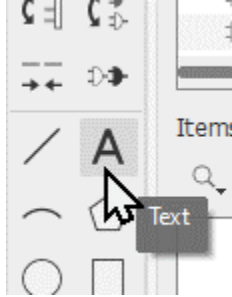
The 'Attributes' section contains a table with the following data:

Name	Value
SPICEPREFIX	R
POPULARITY	48

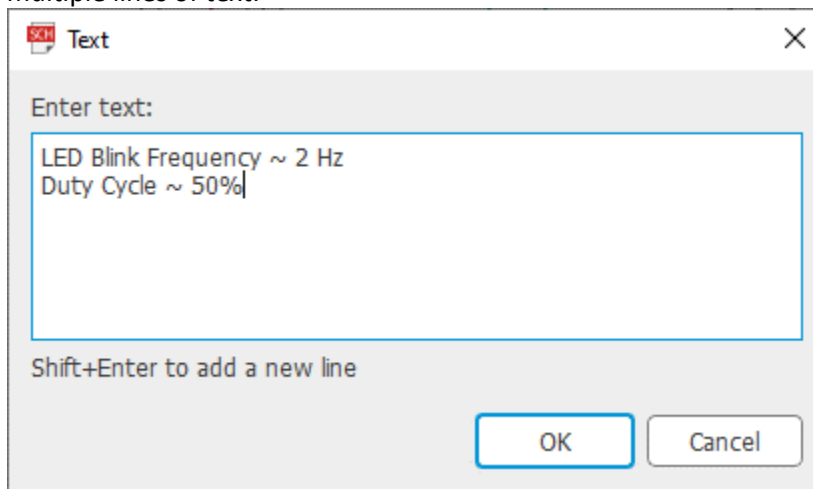
At the bottom of the dialog are three buttons: OK, Cancel, and Apply.

In the **Value** field, give the passive component an appropriate value. Unit suffixes can be specified to reduce the amount of text a value takes up on the schematic, and to increase readability. Note that the LED has a value of **RED**. Feel free to be a super special person and change it to something else.

Finally, let's add a text block to give anyone looking at our schematic some more information about our 555 timer circuit. Use the **Text** tool in the sidebar for this (**text** in the command line also works).

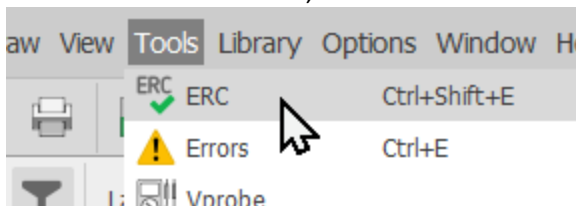


Type out some information on the blink timing and duty cycle of our circuit. Use Shift+Enter to type out multiple lines of text.



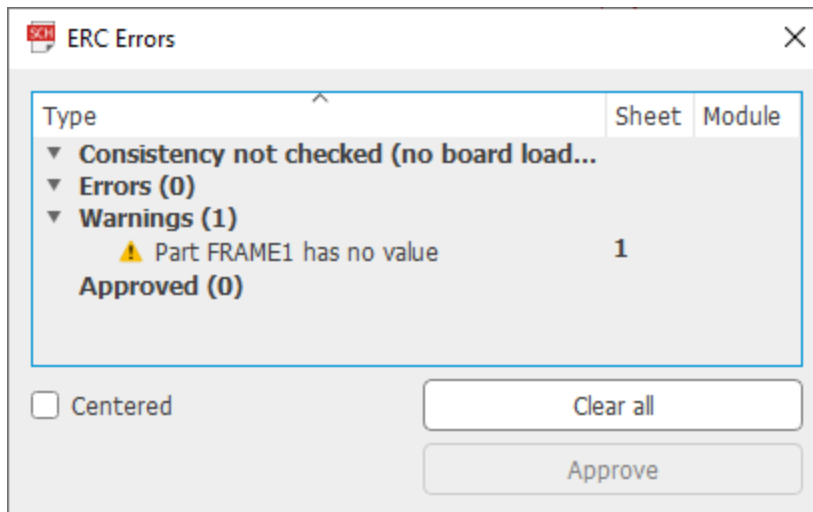
Great! We're done with our schematic. Let's run an ERC.

In the Schematic window, select **Tools > ERC**.



If you get any errors, you can click on them, and EAGLE will point out where on your schematic the errors are occurring. In the vast majority of cases, you must fix all errors reported by ERC.

If you get any warnings, address them on a case-by-case basis. The only warning that should pop up is a warning about our frame not having its Value field filled out. This is fine.



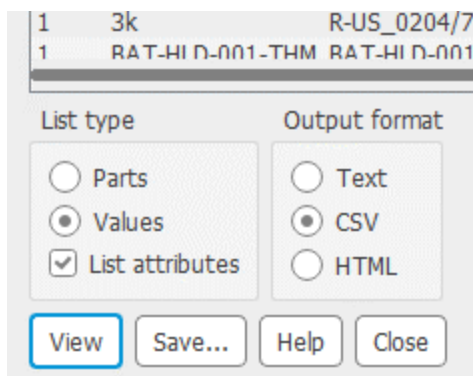
EAGLE Schematic Tools

Let's generate a BOM of our circuit. It's useful in case we ever want to manufacture our project.

In the Schematic window, select **File > Export > BOM**.

Generally for purchasing purposes, listing items by their Values instead of their Parts is more helpful, since each unique component is put on a separate row, along with their total quantity used in the entire schematic. However, listing by parts can be useful for finding individual components if you're manufacturing the board on your own.

Outputting the BOM table as a CSV is also helpful for quick imports into spreadsheet programs such as Microsoft Excel.



Save the BOM however you like, and include it in your journal.

We'll also export our schematic. You can choose to do so as either a PDF or an image.

PDF Export (printing):

1. File > Print
2. For the **Printer**, select **Print to File (PDF)**

3. Select **All** in the **Schematic sheets** section if you have multiple sheets in your schematic
4. Make sure the paper size matches the size you used for your title block frame (A4 in our case)
5. Adjust other attributes as desired, and hit OK to print
6. The PDF will be found in your EAGLE project directory

Image Export (current schematic sheet only):

1. File > Export > Image
2. Specify a filename for your image
3. **Area** should be set to **Full** unless you only want to capture what you currently see in your Schematic window viewport
4. Adjust other attributes as desired, and hit OK to print
5. The image will be found in your EAGLE project directory

Include a copy of your finished schematic in your journal.

Make sure you save a copy of your project for the next lab on PCB layout!
It continues on from the end of this lab.